HOMEWORK-2

1.Integer array using pointer arithmetic

```
#include <stdio.h>


int main()
{
   int arr[100],n;
   int *p;
   int i;
printf("Enter the no of elements:");
scanf("%d",&n);
printf("Enter the array elements:");
for(i=0;i<n;i++)
scanf("%d",&arr[i]);
   p = arr;


   printf("Array elements are:\n");


   for(i = 0; i < n; i++)
   {
      printf("%d\n ", *(p+i));


   }


   for(i = 0; i < n; i++)
   {
```

```
        *(p+i) = *(p+i) + 5;



    }

    printf("Modified array:\n");

    for(i = 0; i < n; i++)
    {
        printf("%d\n ", *(p+i));
    }

    return 0;
}
```

OUTPUT

Enter the no of elements:5

Enter the array elements:1 2 3 4 5

Array elements are:

1

2

3

4

5

Modified array:

6

7

8

9

10

----------2. Calculate string 2.length using pointer

```
#include <stdio.h>


int main()
{
   char str[100];
   char *p;
   int length = 0;


   printf("Enter a string:");
   fgets(str,sizeof(str),stdin);


   p = str;


   while(*p != '\0')
   {
   if(*p != '\n' && *p!=' ')
      length++;
      p++;


   }
```

```
    printf("Length of the string = %d", length);


    return 0;
}
```

OUTPUT

Enter a string:Good morning

Length of the string = 11

---

3. Function to calculate sum and product using pointers

```
#include <stdio.h>


void calculate(int a, int b, int *sum, int *product)
{
    *sum = a + b;
    *product = a * b;
}


int main()
{
    int a, b;
    int sum, product;


    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
```

```
    calculate(a, b, &sum, &product);

    printf("Sum = %d\n", sum);
    printf("Product = %d\n", product);

    return 0;
}
```

OUTPUT

Enter two numbers: 17

8

Sum = 25

Product = 136

4.. Array of structures using dynamic memory

```
#include <stdio.h>
#include <stdlib.h>

struct Student
{
    int id;
    char name[30];
    float mark;
};

int main()
{
```

```
struct Student *ptr;
int n, i;

printf("Enter number of students: ");
scanf("%d", &n);

ptr = (struct Student *)malloc(n * sizeof(struct Student));

if(ptr == NULL)
{
   printf("Memory allocation failed.");
   return 1;
}

for(i = 0; i < n; i++)
{
   printf("\nEnter details of student %d\n", i + 1);

   printf("ID: ");
   scanf("%d", &(ptr+i)->id);

   printf("Name: ");
   scanf("%s", (ptr+i)->name);

   printf("Mark: ");
   scanf("%f", &(ptr+i)->mark);
```

```
    }

    printf("\nStudent Details:\n");

    for(i = 0; i < n; i++)
    {
        printf("\nID: %d", (ptr+i)->id);
        printf("\nName: %s", (ptr+i)->name);
        printf("\nMark: %.2f\n", (ptr+i)->mark);
    }

    free(ptr);

    return 0;
}
```

OUTPUT

Enter number of students: 3

Enter details of student 1

ID: 23

Name: Rahul.V

Mark: 99

Enter details of student 2

ID: 34

Name: Harish.K

Mark: 88

Enter details of student 3

ID: 69

Name: Pavtharani.K.S

Mark: 90

Student Details:

ID: 23

Name: Rahul.V

Mark: 99.00

ID: 34

Name: Harish.K

Mark: 88.00

ID: 69

Name: Pavtharani.K.S

Mark: 90.00

5.Function pointer

```
#include <stdio.h>
```

```
void add(int a, int b)
{
   printf("Addition = %d\n", a + b);
}

void multiply(int a, int b)
{
   printf("Multiplication = %d\n", a * b);
}

int main()
{
   int x,y;
   printf("Enter two numbers:");
   scanf("%d %d",&x,&y);
   void (*fp)(int, int);

   fp = add;
   fp(x, y);

   fp = multiply;
   fp(x, y);

   return 0;
}
```

OUTPUT

Enter two numbers:7 9

Addition = 16

Multiplication = 63

---

6. Pointer and double pointer

```
#include <stdio.h>


int main()
{
   int x ;
   int *p;
   int **q;
printf("Enter the number:");
scanf("%d",&x);


   p = &x;
   q = &p;


   printf("Value of x      = %d\n", x);
   printf("Address of x    = %p\n\n", (void *)&x);


   printf("Value of p      = %p\n", (void *)p);
   printf("Address of p    = %p\n", (void *)&p);
   printf("Value pointed by p = %d\n\n", *p);


   printf("Value of q      = %p\n", (void *)q);
```

```
    printf("Address of q    = %p\n", (void *)&q);
    printf("Value pointed by q = %p\n", (void *)*q);
    printf("Value of **q    = %d\n", **q);

    return 0;
}
```

OUTPUT

Enter the number:4

Value of x   = 4

Address of x   = 0x7ffcacf7993c

Value of p   = 0x7ffcacf7993c

Address of p   = 0x7ffcacf79930

Value pointed by p = 4

Value of q   = 0x7ffcacf79930

Address of q   = 0x7ffcacf79928

Value pointed by q = 0x7ffcacf7993c

Value of **q   = 4

7.Selection Sort using pointers + user input

```
#include<stdio.h>
int main()
{
    float arr[100];
    float *p = arr;
```

```
float temp;
int n, i, j, min;
printf("Enter number of elements: ");
scanf("%d", &n);

printf("Enter %d elements:\n", n);

for(i = 0; i < n; i++)
{
   scanf("%f", p + i);
}

for(i = 0; i < n - 1; i++)
{
   min = i;

   for(j = i + 1; j < n; j++)
   {
      if(*(p + j) < *(p + min))
      {
         min = j;
      }
   }

   temp = *(p + i);
   *(p + i) = *(p + min);
```

```
        *(p + min) = temp;
    }

    printf("Sorted array:\n");

    for(i = 0; i < n; i++)
    {
        printf("%.1f ", *(p + i));
    }

    return 0;
}
```

OUTPUT:

Enter number of elements: 4

Enter 4 elements:

2.3

6.8

9.0

7.8

Sorted array:

2.3 6.8 7.8 9.0

---

8. Student structure and sorting

```
#include <stdio.h>
```

```
struct Student
{
   int id;
   char name[30];
   float grade;
};

int main()
{
   struct Student s[100], temp;
   int n, i, j;

   printf("Enter number of students: ");
   scanf("%d", &n);

   for(i = 0; i < n; i++)
   {
      printf("\nEnter details of student %d\n", i + 1);

      printf("ID: ");
      scanf("%d", &s[i].id);

      printf("Name: ");
      scanf("%s", s[i].name);

      printf("Grade: ");
```

```
    scanf("%f", &s[i].grade);
}



for(i = 0; i < n - 1; i++)
{
    for(j = i + 1; j < n; j++)
    {
        if(s[i].grade > s[j].grade)
        {
            temp = s[i];
            s[i] = s[j];
            s[j] = temp;
        }
    }
}

printf("\nStudents sorted based on grade:\n");

for(i = 0; i < n; i++)
{
    printf("ID: %d  Name: %s  Grade: %.2f\n",
           s[i].id, s[i].name, s[i].grade);
}

return 0;
```

}

OUTPUT:

Enter number of students: 4

Enter details of student 1

ID: 67

Name: Pavatharani

Grade: 78

Enter details of student 2

ID: 25

Name: Rahul

Grade: 67

Enter details of student 3

ID: 43

Name: Harish

Grade: 99

Enter details of student 4

ID: 89

Name: Thirumurugan

Grade: 76

Students sorted based on grade:

ID: 25 Name: Rahul Grade: 67.00

ID: 89 Name: Thirumurugan Grade: 76.00

ID: 67 Name: Pavatharani Grade: 78.00

ID: 43 Name: Harish Grade: 99.00

9.Radix Sort for negative numbers

```
#include <stdio.h>
#include <stdlib.h>

void radixSort(int arr[], int n)
{
   int output[100];
   int count[10];
   int max, exp, i;

   if (n == 0)
      return;

   max = arr[0];

   for (i = 1; i < n; i++)
   {
      if (arr[i] > max)
         max = arr[i];
   }

   for (exp = 1; max / exp > 0; exp *= 10)
```

```
  {
    for (i = 0; i < 10; i++)
      count[i] = 0;

    for (i = 0; i < n; i++)
      count[(arr[i] / exp) % 10]++;

    for (i = 1; i < 10; i++)
      count[i] = count[i] + count[i - 1];

    for (i = n - 1; i >= 0; i--)
    {
      output[count[(arr[i] / exp) % 10] - 1] = arr[i];
      count[(arr[i] / exp) % 10]--;
    }

    for (i = 0; i < n; i++)
      arr[i] = output[i];
  }
}

int main()
{
  int arr[100];
  int negative[100], positive[100];
  int n, i;
```

```
int negCount = 0, posCount = 0;

printf("Enter number of elements: ");
scanf("%d", &n);

printf("Enter %d numbers:\n", n);

for (i = 0; i < n; i++)
{
   scanf("%d", &arr[i]);

   if (arr[i] < 0)
   {
      negative[negCount] = -arr[i];
      negCount++;
   }
   else
   {
      positive[posCount] = arr[i];
      posCount++;
   }
}

radixSort(negative, negCount);

radixSort(positive, posCount);
```

```
    printf("Sorted array:\n");



    for (i = negCount - 1; i >= 0; i--)
    {
        printf("%d ", -negative[i]);
    }



    for (i = 0; i < posCount; i++)
    {
        printf("%d ", positive[i]);
    }


    return 0;
}
```

OUTPUT:

Enter number of elements: 5

Enter 5 numbers:

-7. Copy one array to another using pointers

This

9

8

-3

2

Sorted array:

-7 -3 2 8 9

10.Copy one array to another using pointers

```
#include <stdio.h>


int main()
{
   int a[100], b[100];
   int *p, *q;
   int n, i;


   printf("Enter number of elements: ");
   scanf("%d", &n);


   printf("Enter array elements:\n");


   for(i = 0; i < n; i++)
   {
      scanf("%d", &a[i]);
   }


   p = a;
   q = b;
   for(i = 0; i < n; i++)
   {
```

```
        *(q + i) = *(p + i);
    }

    printf("Copied array:\n");

    for(i = 0; i < n; i++)
    {
        printf("%d ", *(q + i));
    }

    return 0;
}
```

OUTPUT:

Enter number of elements: 3

Enter array elements:

23

45

7

Copied array:

23 45 7
